

DevOps Engineer

Step 4: Learn Kubernetes

CareerCompass — Detailed Learning Notes

What this step is about

Learn container orchestration after Docker fundamentals.

Core Concepts

- Understand Pods, Deployments, Services, ConfigMaps, Secrets, namespaces and declarative desired state.
- Learn kubectl, YAML manifests, service discovery, scaling and rolling updates.
- Understand readiness/liveness probes and resource requests/limits.
- Use local clusters such as Minikube or kind for learning before managing production clusters.

Skills You Should Be Able To Demonstrate

- Create a local cluster.
- Deploy an application.
- Expose it with a Service.
- Scale replicas.
- Perform a rolling update and inspect events/logs.

Hands-on Project / Practice

- Deploy a two-service application locally on Kubernetes.
- Write Deployment and Service manifests and document troubleshooting commands.

Recommended References

- Kubernetes Documentation — <https://kubernetes.io/docs/>
- Kubernetes Course — https://www.youtube.com/watch?v=d6WC5n9G_sM